25901713 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCIAM100 - INTELLIGENT AUTOMATION TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Which of the following is not an application of artificial intelligence? [U] / [CO1]

   a). Self-driving cars b). Chatboats c). Face Recognition Systems d). Data Base Management System

2. How do you install Jupyter Notebook? [U] / [CO2]

   a). install jupyter b). pip install notebook c). python install jupyter d). notebook install

3. What happens if too many clusters are formed in hierarchical clustering? [U] / [CO3]

   a). The accuracy increases b). The computational cost decreases

   c). The clusters become more meaningful d). The model loses interpretability

4. Which of the following is an application of Machine Learning in medical treatment? [U] / [CO1]

a). Image recognition for disease diagnosis	b). Developing new drugs through genetic algorithms	c). Personalized treatment plans based on patient data	d). All of the above

5. Which method is used for regression in KNN? [U] / [CO3]

a). Finding the mode of K nearest neighbors	b). Applying decision trees
c). Taking the average of K nearest neighbors	d). Using a neural network

6. What does unsupervised learning aim to do? [U] / [CO1]

a). Predict the output based on input data	b). Find hidden patterns or intrinsic structures in input data	c). Learn from reward-based feedback	d). Apply past learning to new situations

7. What happens when gradient descent reaches a saddle point? [U] / [CO2]

a). The gradient becomes zero, and progress slows down	b). The algorithm diverges	c). The function value keeps decreasing indefinitely	d). The learning rate increases

8. Which of the following is a key characteristic of unsupervised learning? [U] / [CO3]

a). It requires labeled data for training	b). It uses reinforcement signals
c). It is only used for classification tasks	d). It works without labeled data

9. What does the regression line represent in linear regression? [U] / [CO3]

a). Boundary line for classification	b). Average of all data points
c). Best fit line for predicting output	d). A random line through the data

10. Which of the following is a subset of Artificial Intelligence? [U] / [CO1]

a). Deep Learning	b). Natural Language Processing
c). Machine Learning	d). All of the above

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Explain the relationship between AI, Machine Learning, and Deep Learning.	[U] / [CO1]
12. How can you execute a Python script in IDLE?	[U] / [CO2]
13. Define Unsupervised Learning and mention one real-life application.	[A] / [CO3]
14. What is the primary Python library used for importing and exporting data in Python?	[K] / [CO2]
15. Explain how Machine Learning is used in healthcare.	[A] / [CO3]
16. What do you understand by 'Training Neural Networks'?	[U] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Discuss how Import and Export data using python	[U] / [CO2]
18. How does model evaluation help in improving Supervised Learning techniques?	[U] / [CO3]
19. What are the key obstacles encountered when deploying machine learning models?	[An] / [CO1]
20. Write a Python program to demonstrate the use of functions, loops, and conditional statements together.	[A] / [CO2]

[10x2 = 20]